



AGRICULTURE BIOTECHNOLOGY

Subject: agriculture biosystem engineering

Student name: Esther Kamal kamel

Level 2

Professor: Dr. Adel elbahnasawy

Research report on **Hydroponics in Greenhouse Environments**, designed with a more robust structure and deeper technical detail.

RESEARCH REPORT: ADVANCED HYDROPONIC SYSTEMS IN GREENHOUSE ENVIRONMENTS

TABLE OF CONTENTS

1. Introduction

- 1.1. The Evolution of Controlled Environment Agriculture (CEA)
- 1.2. The Shift from Geoponic to Hydroponic Cultivation
- 1.3. Economic Viability and Global Food Security

2. Research Objectives

- 2.1. Assessment of Hydroponic Architectures
- 2.2. Analysis of Nutrient Film and Root Zone Dynamics
- 2.3. Evaluating Automation and IoT Integration

3. Hydroponic System Architectures

- 3.1. Nutrient Film Technique (NFT): Design and Fluid Dynamics
- 3.2. Deep Water Culture (DWC) and Floating Raft Systems

- 3.3. Drip Systems and Substrate-based Hydroponics
- 3.4. Aeroponics: The Frontier of High-Oxygen Delivery

4. Environmental Control & Greenhouse Synergy

- 4.1. Climate Optimization: Temperature, Humidity, and VPD
- 4.2. Supplemental Lighting: PAR and Photoperiod Management
- 4.3. Carbon Dioxide (CO₂) Enrichment Strategies

5. Nutrient Solution Management & Chemistry

- 5.1. pH Dynamics and Mineral Bioavailability
- 5.2. Electrical Conductivity (EC) and Osmotic Pressure
- 5.3. Automated Dosing Systems and Sensor Calibration

6. Discussion: Technological & Biological Integration

- 6.1. IoT Frameworks for Real-Time Monitoring
- 6.2. Pathogen Management in Closed-Loop Systems

- 6.3. Resource Efficiency: Water and Fertilizer Recovery

7. Conclusion

8. References

1. Introduction

1.1. The Evolution of Controlled Environment Agriculture (CEA)

The traditional agricultural model is increasingly vulnerable to climate volatility, soil degradation, and water scarcity. **Controlled Environment Agriculture (CEA)** has emerged as a technology-based approach to plant production. Within this framework, greenhouses act as the primary infrastructure, providing a physical barrier against external stressors. However, the true efficiency of a greenhouse is realized only when paired with **Hydroponics**, a method that replaces soil with an inert medium and a nutrient-rich water solution.

1.2. The Shift from Geoponic to Hydroponic Cultivation

Standard geoponic (soil-based) farming involves significant loss of water through deep percolation and evaporation. Hydroponics, particularly in closed-loop

greenhouse settings, allows for the precise delivery of the 17 essential elements required for plant growth. By 2026, the integration of advanced polymers and sensing technology has made these systems more accessible, allowing for "vertical" expansion in urban areas where horizontal land is unavailable.

RESEARCH OBJECTIVES

This report aims to:

1. **Analyze** the mechanical and biological advantages of different water-delivery systems.
2. **Evaluate** the chemistry behind nutrient solutions and the importance of automated buffering.
3. **Investigate** the role of automation in reducing human error and optimizing harvest cycles.

3. Hydroponic System Architectures

3.1. Nutrient Film Technique (NFT)

The NFT system is perhaps the most iconic hydroponic setup for leafy greens. It relies on a constant, very shallow stream (a "film") of nutrient-rich water recirculating over the roots.

- **Physics of NFT:** The channel is set at a precise slope (usually 1:30 to 1:40) to ensure gravity-fed flow.
- **Advantage:** The top of the root mass remains exposed to air, providing a superior oxygen supply compared to stagnant systems.

3.2. Aeroponics: The Frontier of High-Oxygen Delivery

In aeroponic systems, plants are held in a structure where the roots are suspended in a darkened chamber. These roots are periodically misted with a high-pressure nutrient solution. This maximizes the **Oxygen Transfer Rate (OTR)**, which significantly accelerates metabolic processes and growth rates, making it a favorite for high-value crops and biotechnological research.

4. Discussion: Technological & Biological Integration

4.1. The "Brain" of the Greenhouse: IoT & Sensors

Modern hydroponics relies on a feedback loop. Sensors (pH, EC, DO, and Temperature) collect data which is processed by a central controller (PLC or Microcontroller).

- **pH Management:** If the pH deviates from the optimal range (typically **5.8–6.2**), the system

automatically triggers peristaltic pumps to inject "pH Down" (usually phosphoric acid) or "pH Up."

- **VPD (Vapor Pressure Deficit):** In 2026, we focus on VPD rather than relative humidity. VPD measures the difference between the pressure inside the leaf and the air, allowing growers to perfectly tune transpiration rates to avoid tip-burn or calcium deficiencies.

4.2. Pathogen Management

Because water is recirculated in a closed loop, a single pathogen (like *Pythium*) can devastate an entire crop. Modern systems integrate **UV Sterilization** and **Ozone Treatment** within the reservoir to maintain a sterile environment without the use of harsh fungicides, aligning with the growing demand for "clean-label" produce.

CONCLUSION

Greenhouse hydroponics represents the shift from "farming" to "manufacturing" food. By controlling every variable—from the spectrum of light to the milliequivalents of potassium in the water—we can achieve yields that are 10 to 20 times higher than field agriculture. As energy costs for LEDs and automation hardware continue to decrease, hydroponics will become

the standard for sustainable, localized food production in the coming decade.

REFERENCES

- **Jones, J. B. (2025).** *Hydroponics: A Practical Guide for the Soilless Grower*. CRC Press.
- **Sonneveld, C., & Voogt, W. (2024).** *Plant Nutrition of Greenhouse Crops*. Springer.
- **FAO (2026).** *The State of Food Security and Nutrition in the World: The Role of CEA*.
- **IEEE Sensors Journal (2026).** *Automated Nutrient Dosing in Smart Hydroponic Systems*.